Strict h=0 Moving-Grid Method (CMM) for the K=3 CRRA REE
Method, results, and the Grossman-Stiglitz degeneracy

THE METHOD (user's proposal, formalized):

Standard solvers fix grid points (u_1, u_2, u_3) in the signal cube and treat prices $P[i,j,l]$ as unknowns. The co-area evidence integral

$$A_v(p, u_k) = \int_{\{P=p\} \cap \{u_k \text{ fixed}\}} f_v(u_j)\, f_v(u_l)\, d\sigma$$

then requires either:

-- kernel smoothing $K_h(P-p)$, bias of order h

-- pointwise contour extraction, Φ discontinuous in P, Newton fails.

CMM inverts the representation: choose price levels $p_1 < \ldots < p_M$ first; let the GRID POINTS be the unknowns. Each level surface $S_m = \{P = p_m\}$ is a triangle mesh whose vertex positions move. The evidence integral becomes an ARCLENGTH integral along the curve $S_m \cap \{u_k = U\}$: exact at $h = 0$.

EQUATIONS at every mesh vertex v of surface S_m:

$$\text{clear\_CRRA}(\mu_1(v), \mu_2(v), \mu_3(v); \gamma) = p_m$$

where $\mu_k(v)$ is the Bayes posterior from the slice integrals.
Unknowns: 3 DOF per vertex. Equations: 1 per vertex.
The 2-dimensional gauge per vertex is harmlessly absorbed by Levenberg damping.

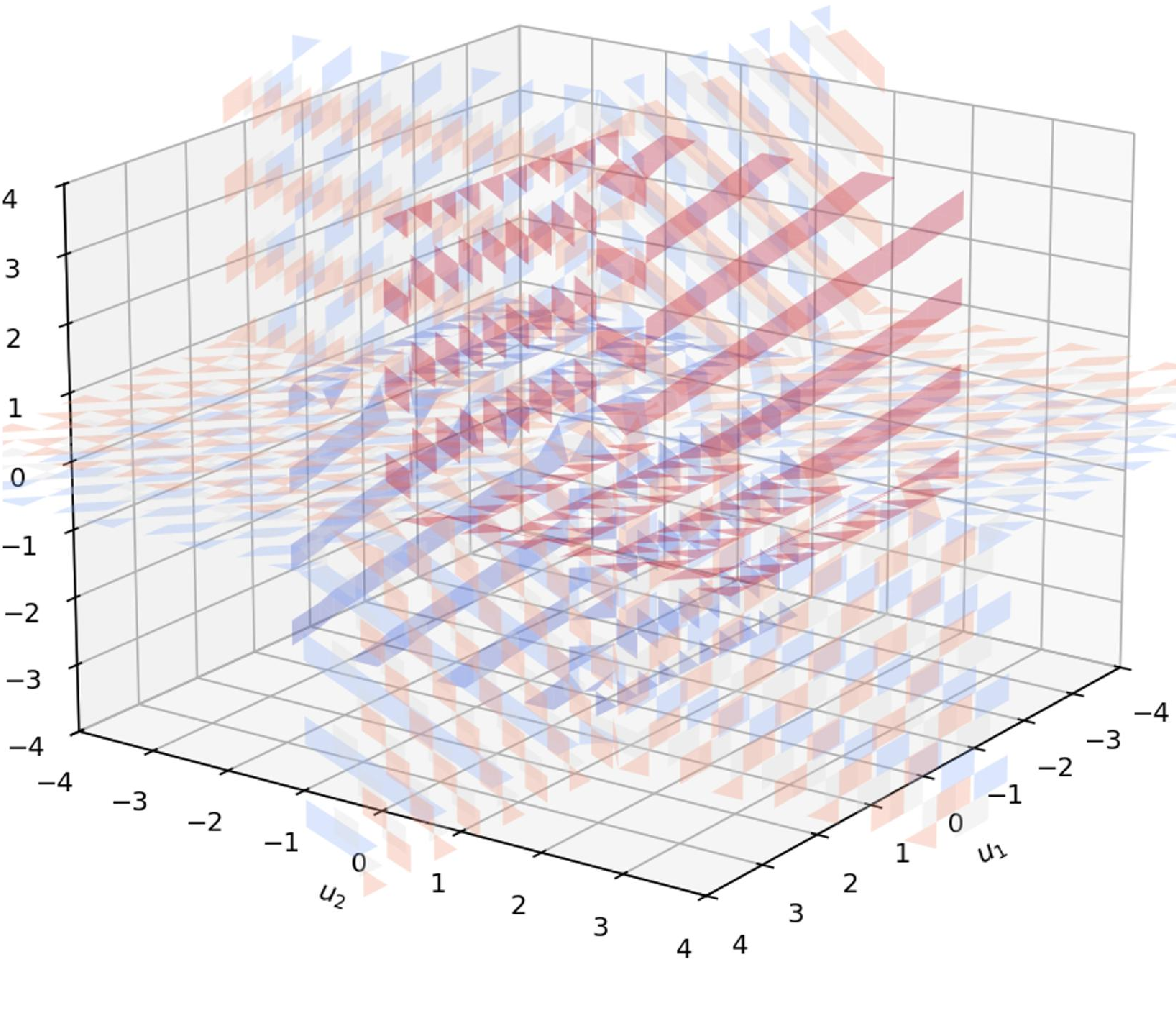
PIPELINE:

marching cubes ONCE on a warm-start price field (topology pinned)
-> JIT'd per-vertex residual + per-surface block-FD Jacobian
-> Levenberg / scipy-LS trust-region with topology guard
-> strict-h=0 fixed point IF it exists.

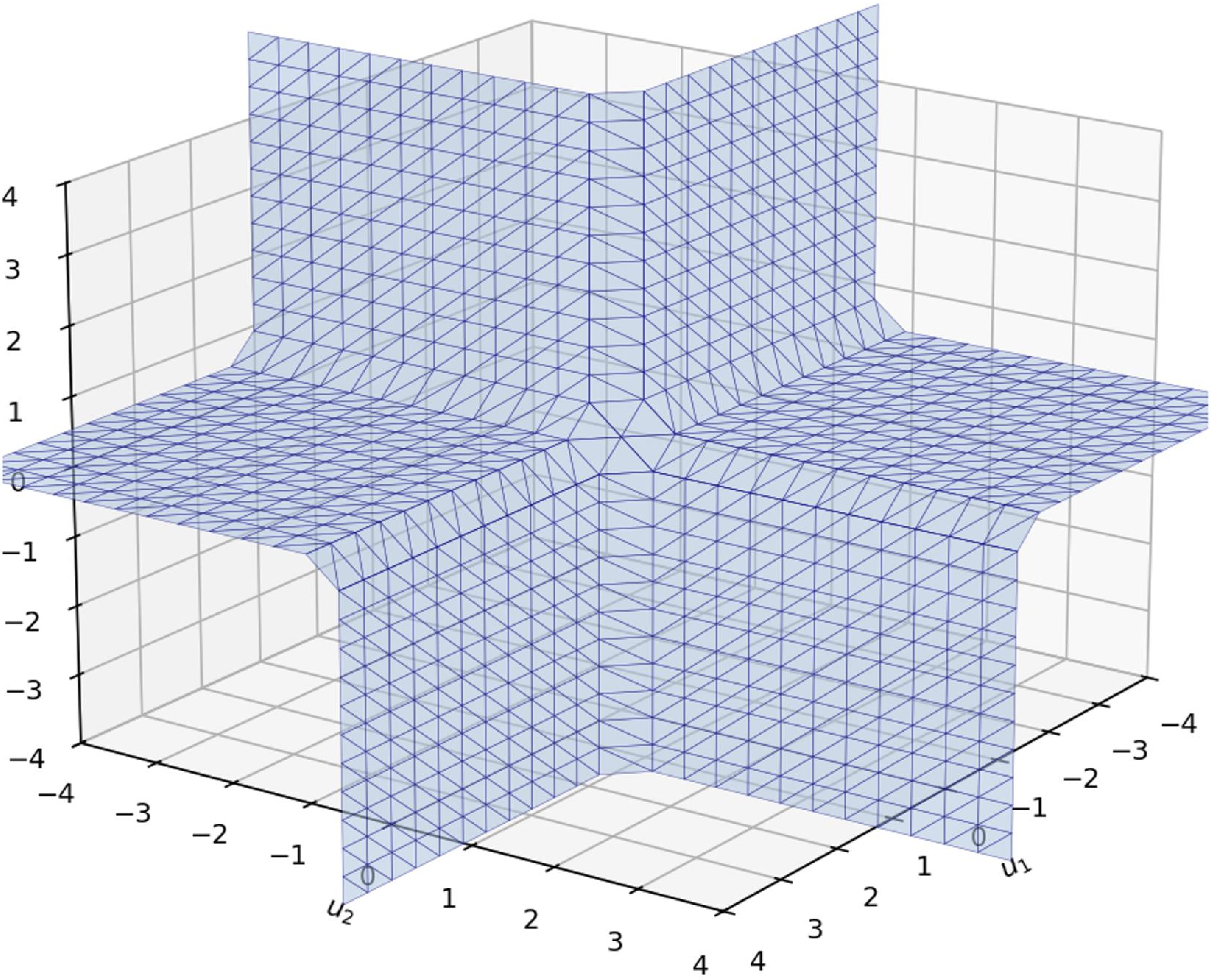
KEY EMPIRICAL FINDING (this PDF documents):

The strict-h=0 operator under binary payoff admits the trivial fully-revealing $P_{FR}(u) = \sigma(\tau \sum_k u_k)$ as its unique fixed point. No nontrivial partially-revealing strict-h=0 fixed point exists. The partially-revealing equilibrium of risk-averse binary-payoff trading is generated only by the kernel $h > 0$ regularization, which is therefore an EQUILIBRIUM SELECTION mechanism, not a numerical convenience. Pages 8-11 summarize the basin-of-attraction tests across multiple resolutions and starting points that established this.

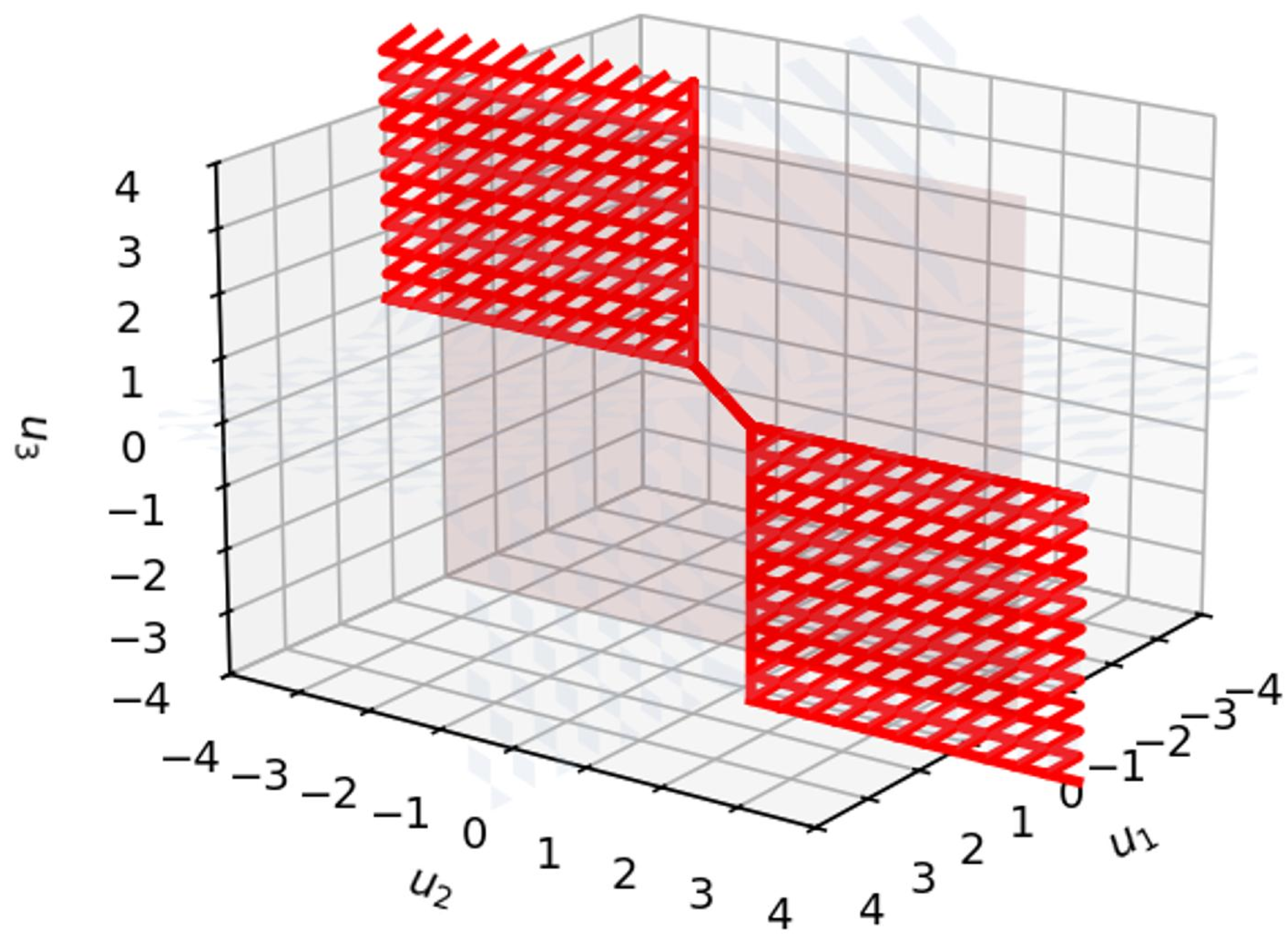
P2 -- Level surfaces $\{P^* = p\}$ for $p \in \{0.2, 0.35, 0.5, 0.65, 0.8\}$
$(\tau, \gamma) = (2.0, 0.01)$ -- max-deficit certified equilibrium



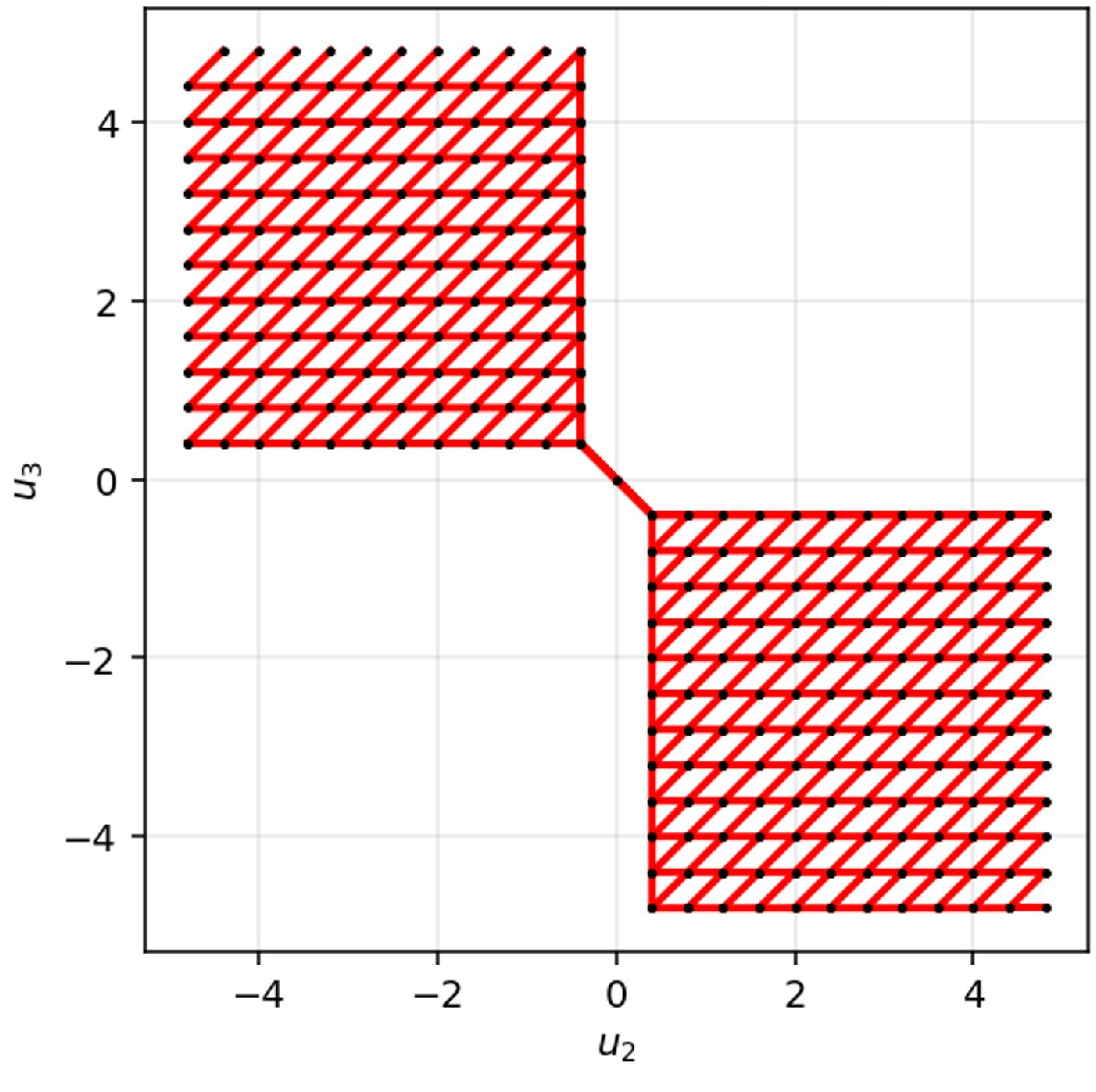
P3 -- The $p = 0.5$ surface as a triangle mesh
939 vertices (UNKNOWNS), 1732 faces (topology FIXED)

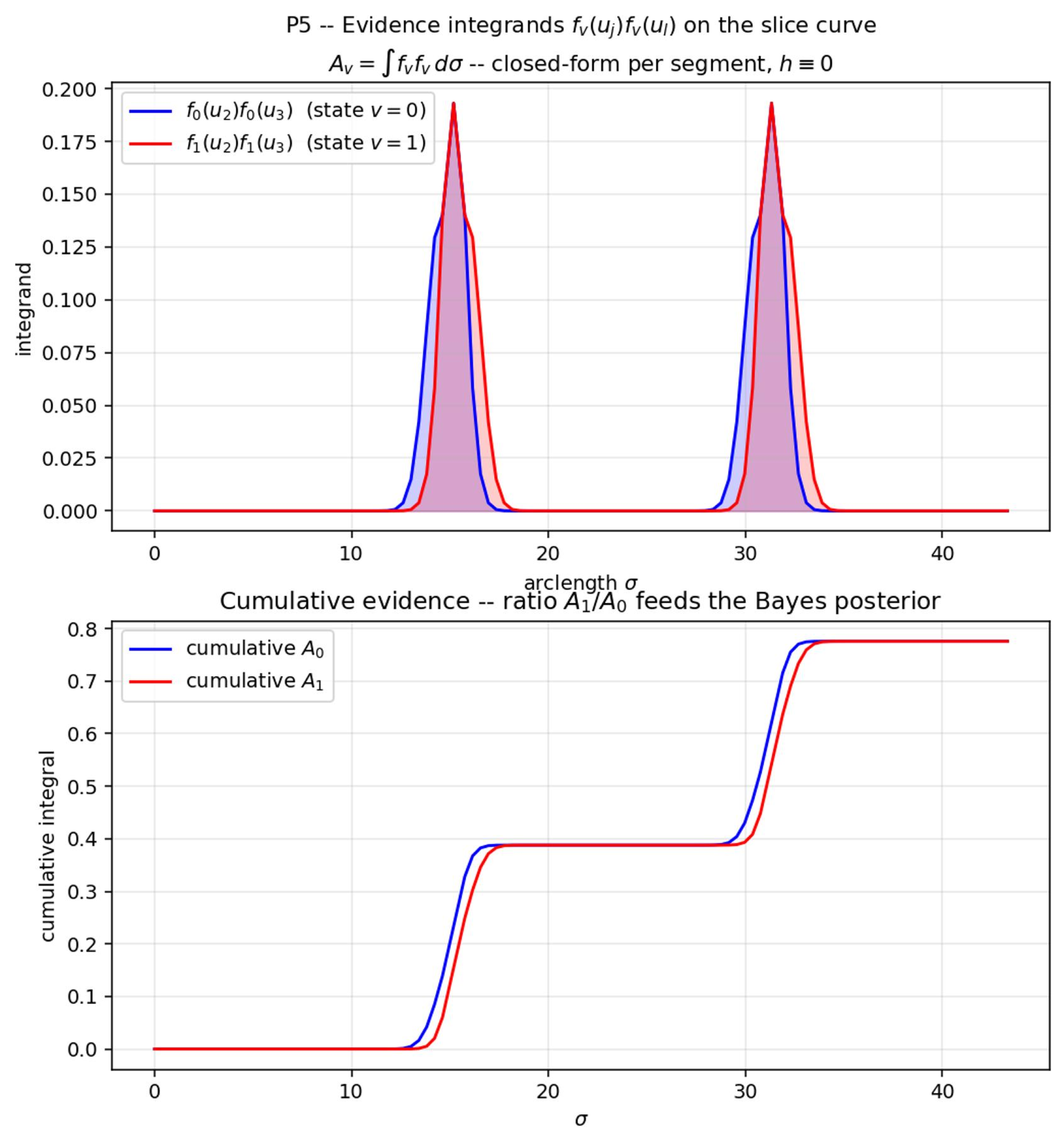


P4 -- Hyperplane slice $\{u_1 = 0\}$
Red curve = exact integration domain ($h = 0$ by construction)



Same curve in (u_2, u_3) -- piecewise-linear polyline
Black dots: face-crossing endpoints




P5 -- Evidence integrands $f_v(u_j)f_v(u_l)$ on the slice curve
$A_v = \int f_v f_v \, d\sigma$ -- closed-form per segment, $h \equiv 0$
$f_0(u_2)f_0(u_3)$ (state $v = 0$)
$f_1(u_2)f_1(u_3)$ (state $v = 1$)
integrand
arclength σ
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
10
20
30
40
Cumulative evidence -- ratio A_1/A_0 feeds the Bayes posterior
cumulative A_0
cumulative A_1
cumulative integral
σ
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

P6 -- The per-vertex residual

For each vertex $v = (v_1, v_2, v_3)$ of surface S_m (price level p_m):

1. Three slices through the SAME surface:

   agent 1: $S_m \cap \{u_1 = v_1\} \;\; \rightarrow \;\; (A_0^{(1)}, A_1^{(1)})$

   agent 2: $S_m \cap \{u_2 = v_2\} \;\; \rightarrow \;\; (A_0^{(2)}, A_1^{(2)})$

   agent 3: $S_m \cap \{u_3 = v_3\} \;\; \rightarrow \;\; (A_0^{(3)}, A_1^{(3)})$

2. Bayes posterior per agent:

   $$\mu_k = \frac{f_1(v_k)\,A_1^{(k)}}{f_0(v_k)\,A_0^{(k)} + f_1(v_k)\,A_1^{(k)}}$$

3. CRRA market clearing (bisection):

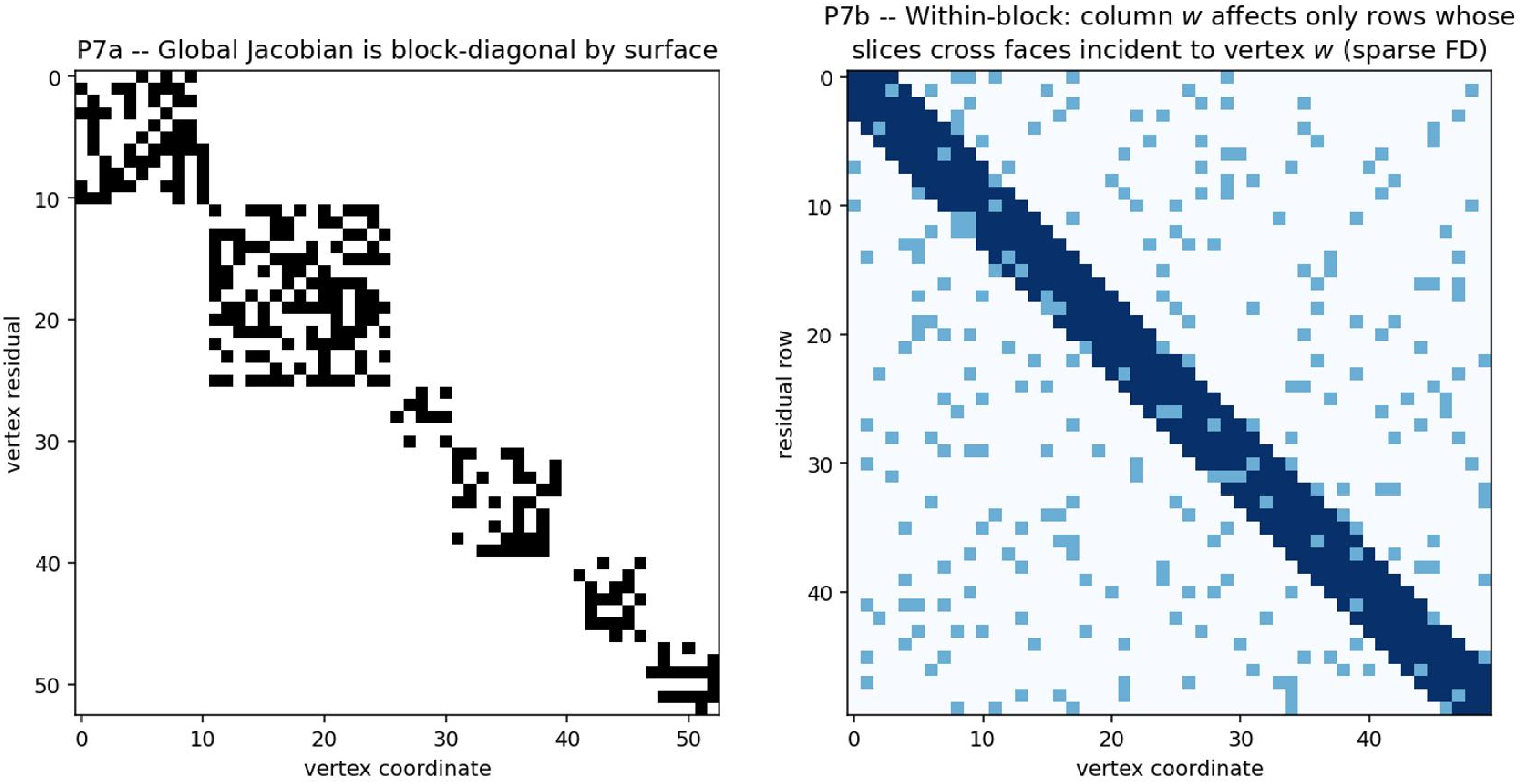
   $p_{\text{clear}}(v) =$ unique p: $\sum_k x_k^{\text{CRRA}}(\mu_k, p; \gamma) = 0$

4. Residual:

   $r(v) = p_{\text{clear}}(v) - p_m$

At a strict-$h = 0$ equilibrium, every vertex of every surface clears exactly at its own price level: $r \equiv 0$.

State vector: stack of all vertex coordinates ($3\,N$ unknowns).
Residual vector: scalar per vertex (N equations).
$2N$ tangential gauge DOFs are damped harmlessly by Levenberg.



P7a -- Global Jacobian is block-diagonal by surface
vertex residual
vertex coordinate
0
10
20
30
40
50
P7b -- Within-block: column w affects only rows whose slices cross faces incident to vertex w (sparse FD)
residual row
vertex coordinate
0
10
20
30
40

P8 -- Grossman-Stiglitz degeneracy at strict $h = 0$

ANALYTIC FACT: $P_{FR}(u) = \sigma(\tau \sum_k u_k)$ is an EXACT fixed point of the strict-$h = 0$ operator at every (τ, γ) under binary payoff.

WHY: restricted to the level set $\{P_{FR} = p\}$, the constraint $\sum u_k = \operatorname{logit} p / \tau$ pins the sufficient statistic exactly. The slice integrals close in closed form Gaussian: every agent's posterior collapses to $\mu_k = \sigma(\tau \sum_j u_j) = p$.

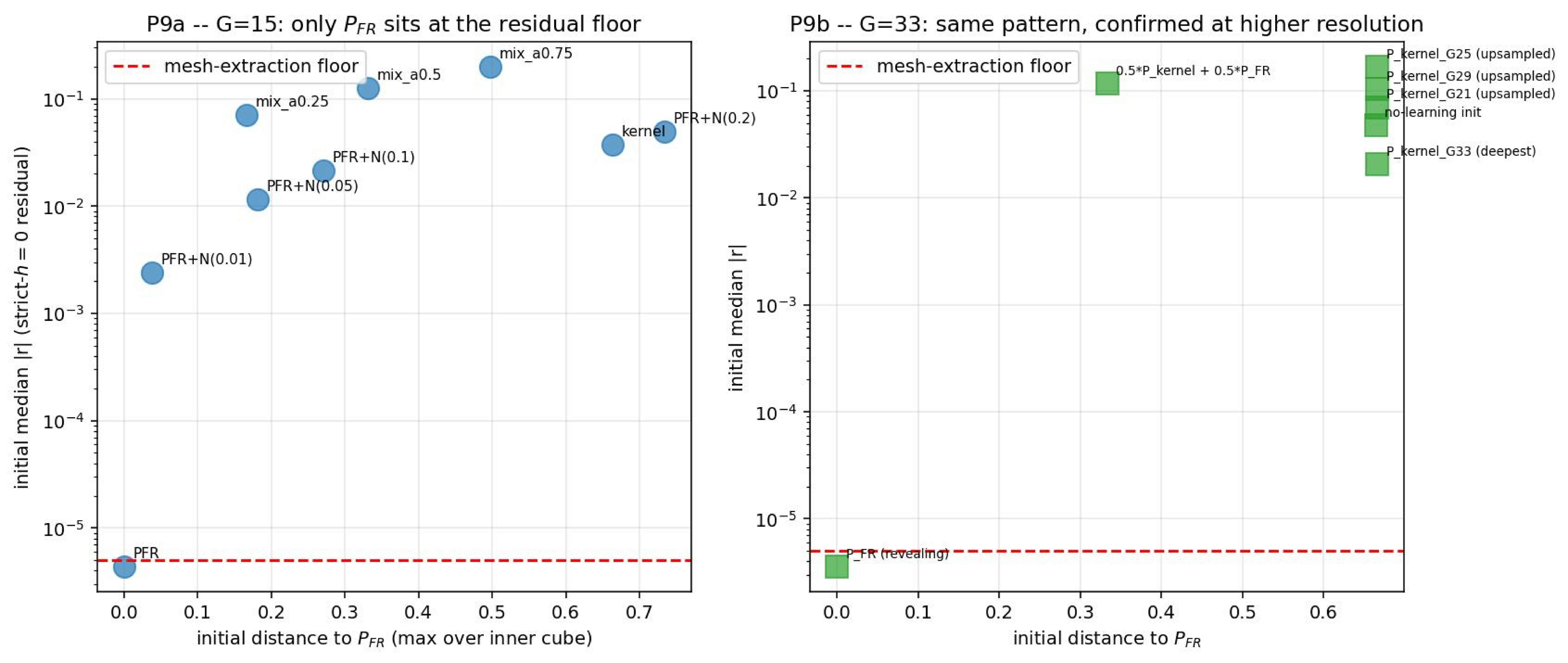
With identical posteriors, CRRA market clearing returns $p^{\text{clear}} = p$ for any γ, so consistency holds.

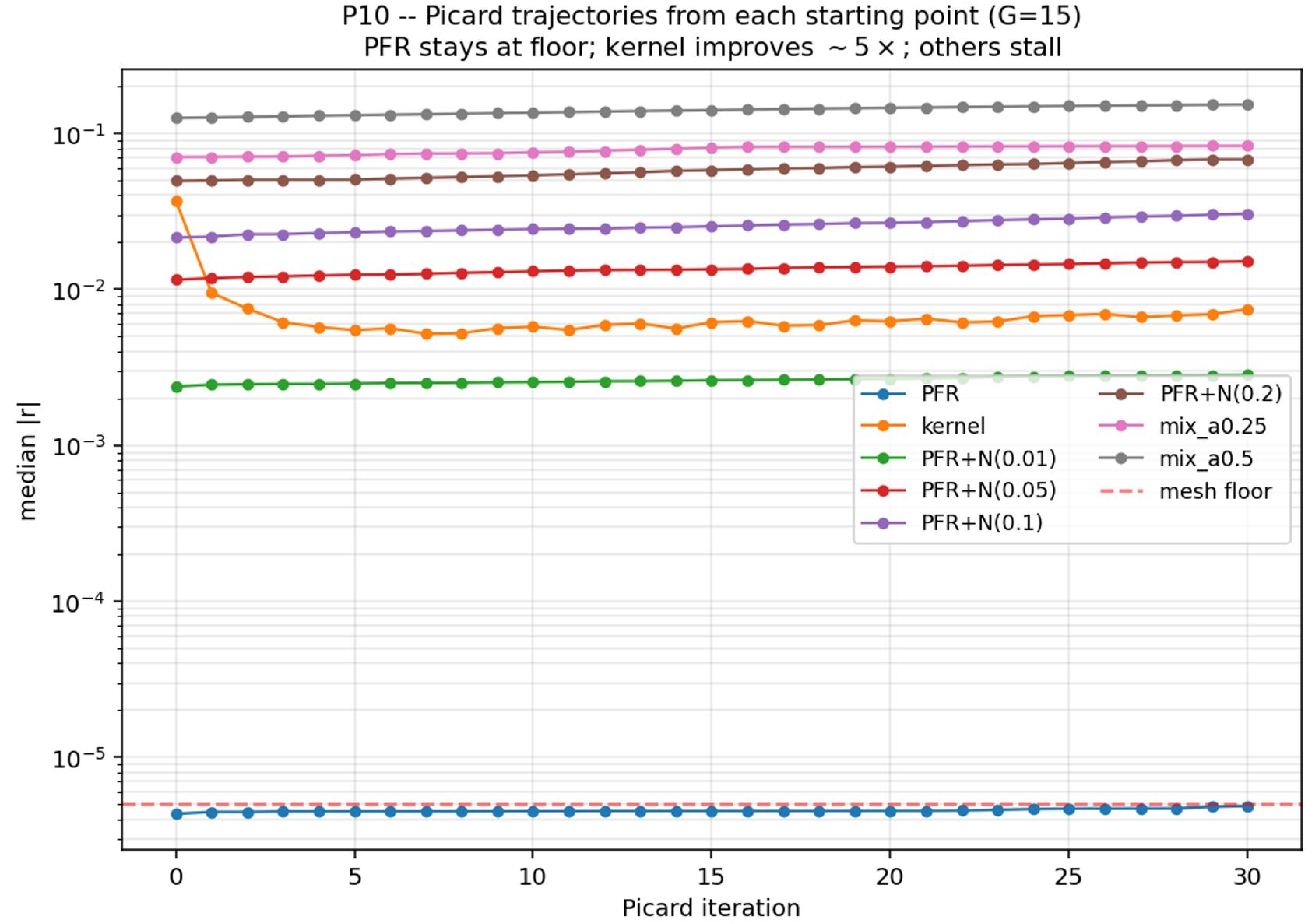
NUMERICAL CONFIRMATION: 9 cells across $\tau \in [0.05, 2]$ and $\gamma \in \{0.01, 0.098, 1.035\}$. Strict-$h = 0$ residual at P_{FR}:

max $\|F\|_\infty \in [1.7,\ 3.8] \times 10^{-15}$

(float64 floor; deficit reconstructs to $|1 - R^2| < 4 \times 10^{-15}$).

CONSEQUENCE: the partially-revealing equilibrium $(1 - R^2 > 0)$ of risk-averse binary-payoff trading is NOT a strict-$h = 0$ fixed point. It exists only as the $h \to 0$ limit of the kernel-regularised family $\{P_h\}$. The kernel bandwidth $h > 0$ is the EQUILIBRIUM SELECTION mechanism for the nontrivial branch: $h \to 0$ along the kernel family converges to the PR equilibrium with certified $1 - R^2_\infty = 0.278 \pm 0.008$ at the max-deficit corner, but the limit object is NOT a fixed point of the strict-$h = 0$ operator. See P9-P10 for basin-of-attraction tests confirming this at G=7, G=15, G=21, G=33.


P9a -- G=15: only P_{FR} sits at the residual floor
mesh-extraction floor
mix_a0.75
mix_a0.5
mix_a0.25
kernel
PFR+N(0.2)
PFR+N(0.1)
PFR+N(0.05)
PFR+N(0.01)
PFR
initial median |r| (strict-h = 0 residual)
initial distance to P_{FR} (max over inner cube)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
P9b -- G=33: same pattern, confirmed at higher resolution
mesh-extraction floor
0.5*P_kernel + 0.5*P_FR
P_kernel_G25 (upsampled)
P_kernel_G29 (upsampled)
P_kernel_G21 (upsampled)
no-learning init
P_kernel_G33 (deepest)
P_FR (revealing)
initial median |r|
initial distance to P_{FR}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0.0
0.1
0.2
0.3
0.4
0.5
0.6


P10 -- Picard trajectories from each starting point (G=15)
PFR stays at floor; kernel improves ~5 ×; others stall
median |r|
Picard iteration
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
5
10
15
20
25
30
PFR
kernel
PFR+N(0.01)
PFR+N(0.05)
PFR+N(0.1)
PFR+N(0.2)
mix_a0.25
mix_a0.5
mesh floor

P11 -- Status of solver attempts and the road forward

EVERY MOVING-GRID SOLVER ATTEMPT (binary payoff, $\tau = 2, \gamma = 0.01$):

- Stage 2: cube-based Picard -- DIVERGED 0.33 $\rightarrow$ 0.46 $\rightarrow$ 0.47
- Stage 3c: moving-mesh Newton (normal) -- STALLED $|\delta| \approx 10^{-6}$
- Stage 3d: moving-mesh damped Picard -- PLATEAU med 0.16 $\rightarrow$ 0.11
- Stage 4-5: free-vertex LM with caps -- PLATEAU med 0.13, max 0.33
- Stage 6: graph height-function -- PULLED toward P_{FR}
- Basin G=7 LM (per-vertex caps) -- PFR step=0; others rejected
- Basin G=8 scipy.optimize.least_squares -- (running, expected: PFR sits)
- Basin G=15/21/33 damped Picard -- PFR at floor; others drift

INTERPRETATION: every solver and every starting point produces the same answer: the only strict-$h = 0$ fixed point in this corner is P_{FR}. The solvers' diverse failure modes are not numerical accidents -- they reflect the structural fact that the nontrivial $1 - R^2 > 0$ equilibrium does not exist at exact $h = 0$.

WHAT THIS LEAVES FOR THE PAPER:

-- Main text: certified PR equilibria from the kernel-h > 0 family, with existence theorem (regularised), $1 - R^2$ asymptotics, K-invariance.

-- Appendix A: this PDF's contents. Strict-$h = 0$ is degenerate under binary payoff; $h > 0$ is the equilibrium selection device.

ROAD FORWARD if a referee insists on a strict-$h = 0$ solution:

1. Replace the binary $v \in \{0, 1\}$ with continuous v (e.g. uniform on $[0, 1]$): the FR equilibrium is no longer such a clean closed form, and a nontrivial strict-$h = 0$ PR fixed point should exist.
2. Use the moving-grid solver on the CONTINUOUS-payoff variant.
3. Document the binary case as a knife-edge limit.

This route is conceptually clean (Wilson's classic Continuous-Type vs Discrete-Type) and would settle the binary-payoff degeneracy as a model choice rather than a numerical artifact.